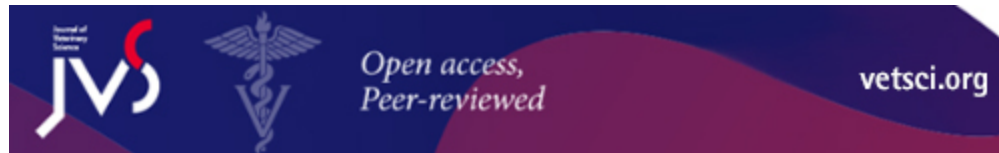


As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)



J Vet Sci. 2026 Mar 26;27(2):e30. doi: [10.4142/jvs.25273](https://doi.org/10.4142/jvs.25273) [↗](#)

Association between chronic kidney disease progression and serum feline pancreatic lipase concentrations in cats

[Jihyun Kim](#)^{1,2}, [Youngmin Yun](#)^{1,✉}

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC13062426 PMID: [41947688](#)

Abstract

Importance

Serum feline pancreatic lipase (fPL) is used to diagnose pancreatitis, but interpretation is challenging because fPL may be affected by non-pancreatic disease, including reduced renal function.

Objective

To evaluate the association between chronic kidney disease (CKD) stage and serum fPL in cats and characterize how renal biomarkers relate to fPL.

Methods

This retrospective study evaluated medical records of cats presented to a secondary referral veterinary hospital between February 2024 and March 2025. Inclusion required same-day testing of complete blood count, serum biochemistry, blood gas analysis, urinalysis, serum fPL, and abdominal ultrasonography. CKD staging followed International Renal Interest Society guidelines using serum creatinine, urine specific gravity (USG), and ultrasonography. Nonparametric group comparisons were performed using the Kruskal–Wallis test, and associations between fPL and renal biomarkers were assessed using Spearman’s correlation.

Results

fPL increased significantly with advancing CKD stage ($p < 0.001$). Furthermore, fPL was positively correlated with blood urea nitrogen ($\rho = 0.516, p < 0.001$), creatinine ($\rho = 0.459, p < 0.001$), and inorganic phosphorus ($\rho = 0.312, p < 0.001$). Additionally, fPL and USG exhibited an inverse correlation ($\rho = -0.502, p < 0.001$). Ionized calcium showed no correlation with fPL ($\rho = 0.012, p = 0.898$).

Conclusions and Relevance

In this clinical cohort, higher CKD stage was associated with higher serum fPL. Because fPL alone is insufficient to diagnose pancreatitis and may increase with reduced renal function, mild-to-moderate fPL elevations in cats with CKD should be interpreted in the context of clinical signs and pancreatic imaging findings.

Keywords: Feline medicine, renal insufficiency, azotemia, pancreatitis, lipase

INTRODUCTION

Chronic kidney disease (CKD) is one of the most prevalent conditions affecting geriatric cats. Its reported prevalence rates in cats older than 10 years exceed 30%–40% [[1,2,3,4,5](#)]. Common historical and clinical symptoms of CKD include weight loss, polyuria, polydipsia, and persistently low urine specific gravity (USG; < 1.035) [[6,7,8,9](#)]. According to the consensus guidelines of the International Society of Feline Medicine, a clinical diagnosis of feline CKD is established based on the presence of azotemia in conjunction with inadequate concentrated urine [[10](#)].

Feline pancreatitis, once considered an uncommon clinical entity, is now recognized as a frequently occurring condition with a prevalence rate comparable to that observed in humans and

dogs. A 2007 histopathological study involving 115 feline necropsy cases demonstrated evidence of pancreatic inflammation in 67% of the participants; chronic pancreatitis, acute pancreatitis (AP), and acute-on-chronic pancreatitis accounted for 50.4%, 6.1%, and 9.6%, respectively. Furthermore, 45% of clinically healthy cats had histopathological signs of pancreatitis [11]. Serum feline pancreatic lipase (fPL) concentration has since emerged as a sensitive and specific biomarker for the non-invasive diagnosis of feline pancreatitis [12,13,14,15].

In human medicine, a substantial body of evidence indicates that the pancreas is adversely affected by uremia [16,17,18]. A strong association has been documented between end-stage renal disease (ESRD) and AP [19,20]. A nationwide cohort study conducted in Germany that involved more than 70,000 patients with ESRD undergoing hemodialysis reported an annual incidence of AP at 67 cases per 100,000 individuals. This was markedly higher than the 19.7 cases per 100,000 observed in the general population [21]. Similarly, a prospective cohort study involving 2,603 ESRD patients in Taiwan found that the incidence of AP was more than threefold more prevalent among patients with ESRD than in the general population. Several pathophysiological mechanisms have been proposed to explain this relationship. One hypothesis suggests that calcium in the peritoneal dialysate may diffuse through the peritoneum, leading to localized hypercalcemia within the pancreas and further exacerbation of pancreatic dysfunction [22]. Additionally, increased concentrations of gastrointestinal hormones such as cholecystokinin, gastric inhibitory polypeptide, and glucagon that occur in renal insufficiency may stimulate pancreatic enzyme hypersecretion and contribute to pancreatic injury [23].

Despite the recognized association between renal dysfunction and pancreatitis in humans, the relationship between CKD and pancreatitis in cats remains difficult to define. Importantly, diagnosis of feline pancreatitis is challenging and typically relies on integration of clinical signs, laboratory abnormalities, and imaging findings rather than a single biomarker. In addition, pancreatic lipase is cleared in part by the kidneys; therefore, reduced glomerular filtration rate may contribute to higher measured lipase concentrations even in the absence of primary pancreatic inflammation.

Therefore, this study aimed to evaluate how serum fPL concentrations vary across CKD stages and how fPL relates to commonly measured renal biomarkers in cats. We hypothesized that fPL concentrations would show an association with markers of reduced renal function, emphasizing the need for cautious interpretation of fPL in cats with CKD.

METHODS

Ethical approval

This retrospective study involved a review of existing clinical data and did not require experimental interventions; therefore, ethical approval was waived. Written informed consent was obtained from all cat owners prior to diagnostic evaluation and use of clinical data. All clinical procedures were performed under the supervision of licensed veterinarians in accordance with institutional and national guidelines. Patient data were anonymized before analysis to ensure confidentiality.

Study design

This retrospective, observational study examined the electronic medical records of cats presented to the secondary referral veterinary hospital between February 2024 and March 2025.

Inclusion criteria

Cats were eligible for inclusion if they underwent a comprehensive diagnostic evaluation comprising complete blood count (CBC), blood gas analysis, serum biochemistry, measurement of serum fPL concentration, urinalysis, and abdominal ultrasonography on the same day as their clinical examination.

CBC was performed using an automated hematology analyzer (ProCyte Dx; IDEXX Laboratories, USA). Serum biochemical analysis was conducted using an automated dry chemistry analyzer (FUJI DRY-CHEM NX700; Fujifilm, Japan). Urinalysis, including USG measurement, was performed using a refractometer on urine samples obtained by cystocentesis at the time of initial diagnostic evaluation during abdominal ultrasonography, prior to therapeutic interventions. Abdominal ultrasonographic examinations were performed using an ultrasound system (ProSound α 7; Hitachi Aloka Medical, Japan) equipped with a 12-MHz linear transducer. Examinations were conducted as part of routine clinical care by a board-certified veterinary radiologist and typically included evaluation of multiple abdominal organs, including the kidneys and pancreas. However, pancreatic ultrasonographic findings and standardized clinical signs compatible with pancreatitis were not consistently recorded across cases and therefore, were not used to define pancreatitis for the purposes of this analysis. A total of 169 cats met the initial inclusion criteria during the study period.

Exclusion criteria

Fifty-seven cats were excluded based on the following criteria: acute kidney injury secondary to nephrotoxic agents such as non-steroidal anti-inflammatory drugs; infectious etiologies including pyelonephritis, neoplastic diseases, or urinary tract obstruction; administration of immunosuppressive therapy (e.g., prednisolone) owing to potential effects on fPL concentrations and USG; and use of diuretics for the management of hypertrophic cardiomyopathy. A total of 112 cats were included in the final analysis after applying these criteria.

Clinical assessment

The 112 cats were categorized into five groups based on serum creatinine concentration, USG, and renal ultrasonographic findings. The normal group had serum creatinine concentrations of < 1.6 mg/dL, USG of ≥ 1.035 , and no detectable abnormalities on renal ultrasonography. CKD staging was performed according to the 2019 International Renal Interest Society (IRIS) guidelines: stage 1 was characterized by serum creatinine concentrations of < 1.6 mg/dL and USG of < 1.035; stage 2 by serum creatinine concentrations between 1.6 and 2.8 mg/dL; stage 3 by serum creatinine concentrations between 2.9 and 5.0 mg/dL; and stage 4 by serum creatinine concentrations of > 5.0 mg/dL. This operational definition was used to identify early CKD (Stage 1) characterized by inadequate urine concentrating ability without azotemia and represents a study-specific modification of the IRIS staging system, in which USG is used for CKD diagnosis rather than formal staging [10]. Renal ultrasonographic abnormalities considered consistent with CKD included increased cortical echogenicity, loss of corticomedullary distinction, renal contour irregularities, and reduced renal size [24].

Serum fPL concentrations were measured using the Vcheck fPL (Bionote) quantitative immunoassay, which is a fluorescence immunochromatographic assay designed for fPL detection. The assay provides quantitative results within 15 min and has a dynamic measurement range of 1–30 ng/mL. According to the manufacturer's recommendations, values of ≤ 3.5 , 3.6–5.3, and ≥ 5.4 ng/mL were categorized as normal, equivocal, and above the assay cutoff, respectively.

Statistical analysis

Comparisons of serum fPL concentrations and USG across CKD stages were performed using the Kruskal–Wallis test due to the non-parametric distribution of the data. Associations between fPL concentrations and renal biomarkers, including blood urea nitrogen (BUN), creatinine, inorganic phosphorus (IP), ionized calcium (iCa), and USG, were assessed using Spearman's rank correlation coefficients. All p values were two-tailed, and p values of < 0.05 were considered statistically

significant. Statistical analyses were conducted using IBM SPSS Statistics, version 22 (IBM, USA), with the nonparametric tests and correlation modules applied.

RESULTS

Patient characteristics

A total of 112 cats were included in this study and categorized according to their CKD stage based on IRIS guidelines. The distribution of CKD stages was as follows: normal: 42 cats (37.5%); stage 1, 13 cats (11.6%); stage 2, 34 cats (30.4%); stage 3, 9 cats (8.0%); and stage 4, 14 cats (12.5%).

Regarding sex distribution, neutered males were predominant (n = 61, 54.5%), followed by spayed females (n = 45, 40.2%). Intact males and females comprised 2.7% and 2.7% of the population, respectively (n = 3 for both).

Domestic Shorthair cats were the most common (n = 53, 47.3%), followed by Persian (n = 15, 13.4%), Russian Blue (n = 8, 7.1%), and Siamese (n = 7, 6.3%) cats. Other breeds included Scottish Fold, Abyssinian, Turkish Angora, Munchkin, Ragdoll, Bengal, British Shorthair, Norwegian Forest, Maine Coon, and American Shorthair cats, each representing a smaller proportion of the population ([Table 1](#)).

Table 1. Demographic characteristics of 112 cats stratified by CKD stage based on the IRIS guidelines.

CKD IRIS stage	Normal	1	2	3	4	Total (%)
Number of cats (%)	42 (37.5)	13 (11.6)	34 (30.4)	9 (8.0)	14 (12.5)	112 (100.0)
Sex						
Intact male	1	0	0	1	1	3 (2.7)
Neutered male	17	9	20	6	9	61 (54.5)
Intact female	1	0	0	2	0	3 (2.7)
Spayed female	23	4	14	0	4	45 (40.2)
Breed						
Domestic Shorthair	19	7	18	3	6	53 (47.3)
Persian	4	1	8	1	1	15 (13.4)
Russian Blue	1	1	2	2	2	8 (7.1)
Siamese	1	0	3	1	2	7 (6.3)
Scottish Fold	3	1	1	0	1	6 (5.4)
Abyssinian	2	1	0	1	0	4 (3.6)
Turkish Angora	1	1	1	0	1	4 (3.6)
Munchkin	4	0	0	0	0	4 (3.6)
Ragdoll	2	0	1	0	0	3 (2.7)
Bengal	2	0	0	1	0	3 (2.7)
British Shorthair	2	0	0	0	0	2 (1.8)
Norwegian Forest	0	1	0	0	0	1 (0.9)

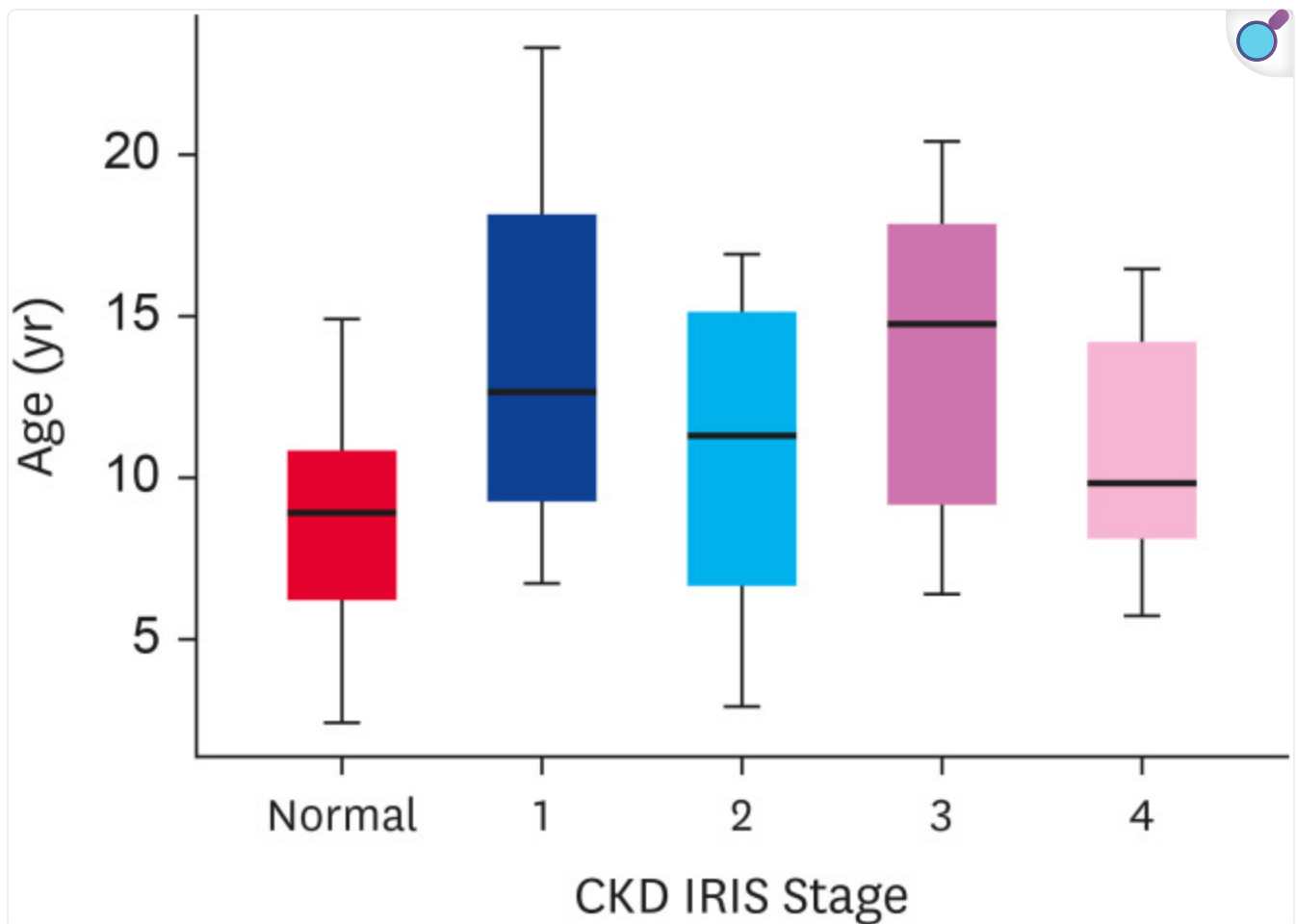
CKD IRIS stage	Normal	1	2	3	4	Total (%)
Maine Coon	1	0	0	0	0	1 (0.9)
American Shorthair	0	0	0	0	1	1 (0.9)
Age						
Median (min–max)	9.0 (2.0–15.1)	13.0 (6.4–23.7)	11.6 (2.6–17.1)	14.8 (5.7–21.7)	10.0 (5.4–18.4)	10.2 (2.0–23.7)

[Open in a new tab](#)

CKD, chronic kidney disease; IRIS, International Renal Interest Society.

The median age of the overall population was 10.2 years (range, 2.0–23.7 years). Median age tended to be higher in cats with CKD, although this trend was not strictly progressive with increasing stage: 9.0 years (2.0–15.1 years) for the normal group, 13.0 years (6.4–23.7 years) for stage 1, 11.6 years (2.6–17.1 years) for stage 2, 14.8 years (5.7–21.7 years) for stage 3, and 10.0 years (5.4–18.4 years) for stage 4 ([Fig. 1](#)).

Fig. 1. Age distribution according to CKD IRIS stage among 112 cats. Box-and-whisker plot showing the distribution of age across different CKD IRIS stages in cats. Each box represents the IQR, the line within each box indicates the median, and the whiskers extend to 1.5 times the IQR. Median age tends to be higher for cats with CKD, but this trend was not strictly progressive with increasing stage: 9.0 years (2.0–15.1 years) for the normal group; 13.0 years (6.4–23.7 years) for stage 1; 11.6 years (2.6–17.1 years) for stage 2; 14.8 years (5.7–21.7 years) for stage 3; and 10.0 years (5.4–18.4 years) for stage 4.



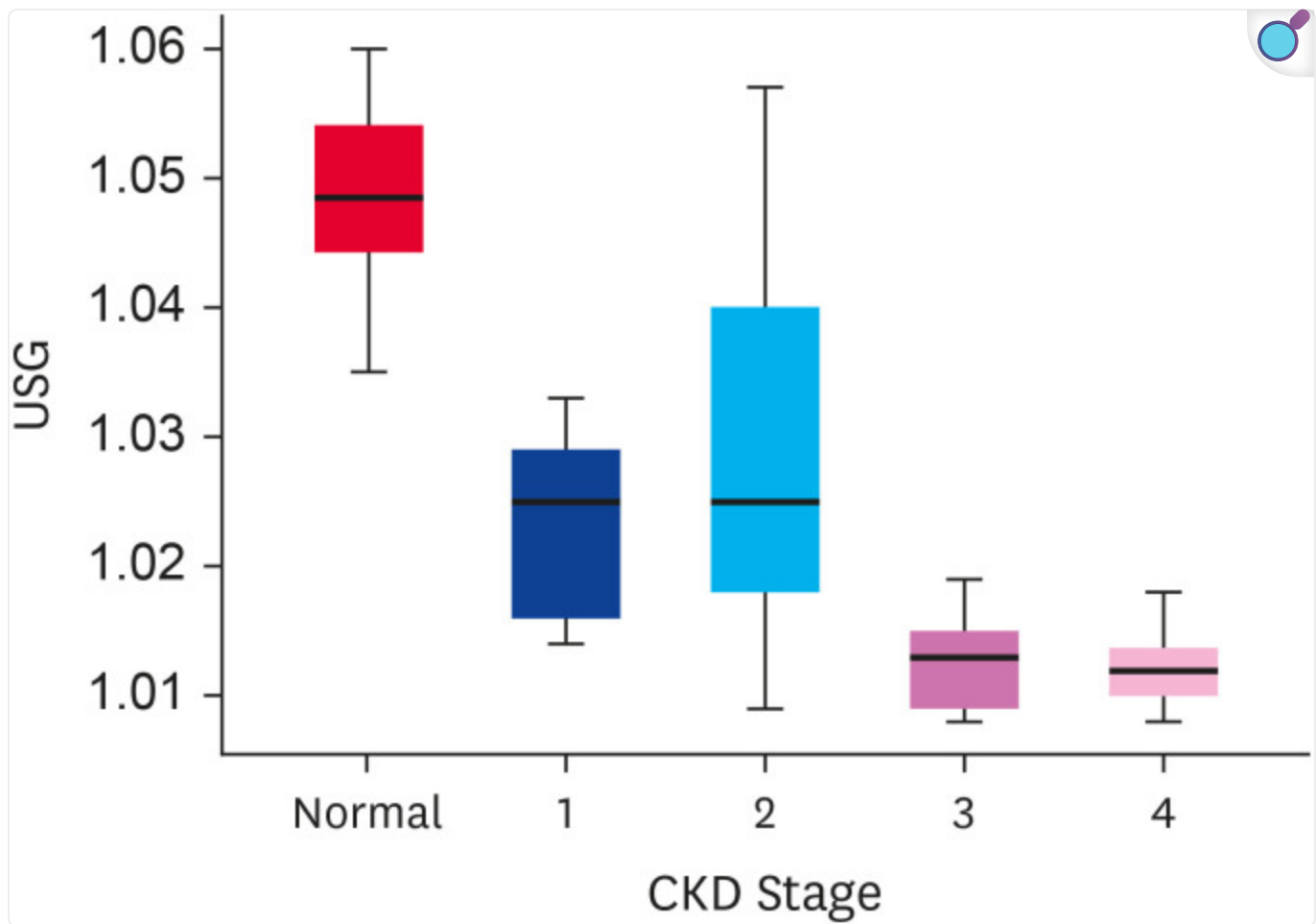
[Open in a new tab](#)

CKD, chronic kidney disease; IRIS, International Renal Interest Society; IQR, interquartile range.

USG was analyzed across CKD IRIS stages for 112 cats. For the normal group, USG values were appropriately high with a median of 1.049 (interquartile range [IQR], 1.044–1.054) and a range of 1.035–1.060, reflecting normal renal concentrating ability.

The median USG values for the CKD stages were as follows: stage 1: 1.025 (IQR, 1.016–1.029; 95% confidence interval [CI], 1.019–1.028); stage 2: 1.025 (IQR, 1.018–1.040; 95% CI, 1.023–1.033); stage 3: 1.013 (IQR, 1.009–1.015; 95% CI, 1.010–1.016); and stage 4: 1.012 (IQR, 1.010–1.014; 95% CI, 1.011–1.014) ([Fig. 2](#)).

Fig. 2. Distribution of USG across CKD stages in cats. Box-and-whisker plot illustrating the distribution of USG values among cats at different CKD IRIS stages.



[Open in a new tab](#)

USG, urine specific gravity; CKD, chronic kidney disease; IRIS, International Renal Interest Society.

A significant decline in USG was observed with increasing CKD severity, although the trend was not strictly progressive. The USG values for cats with stage 1 disease showed a modest decrease relative to those for normal cats, while those with stage 2 disease had a wider distribution with some overlap. In contrast, marked reductions in USG were evident for stages 3 and 4, where values clustered tightly around 1.009–1.015, indicating impaired renal concentrating ability. A Kruskal–Wallis H test revealed that the differences in USG values across CKD stages were statistically significant ($H = 80.53, p < 0.001$).

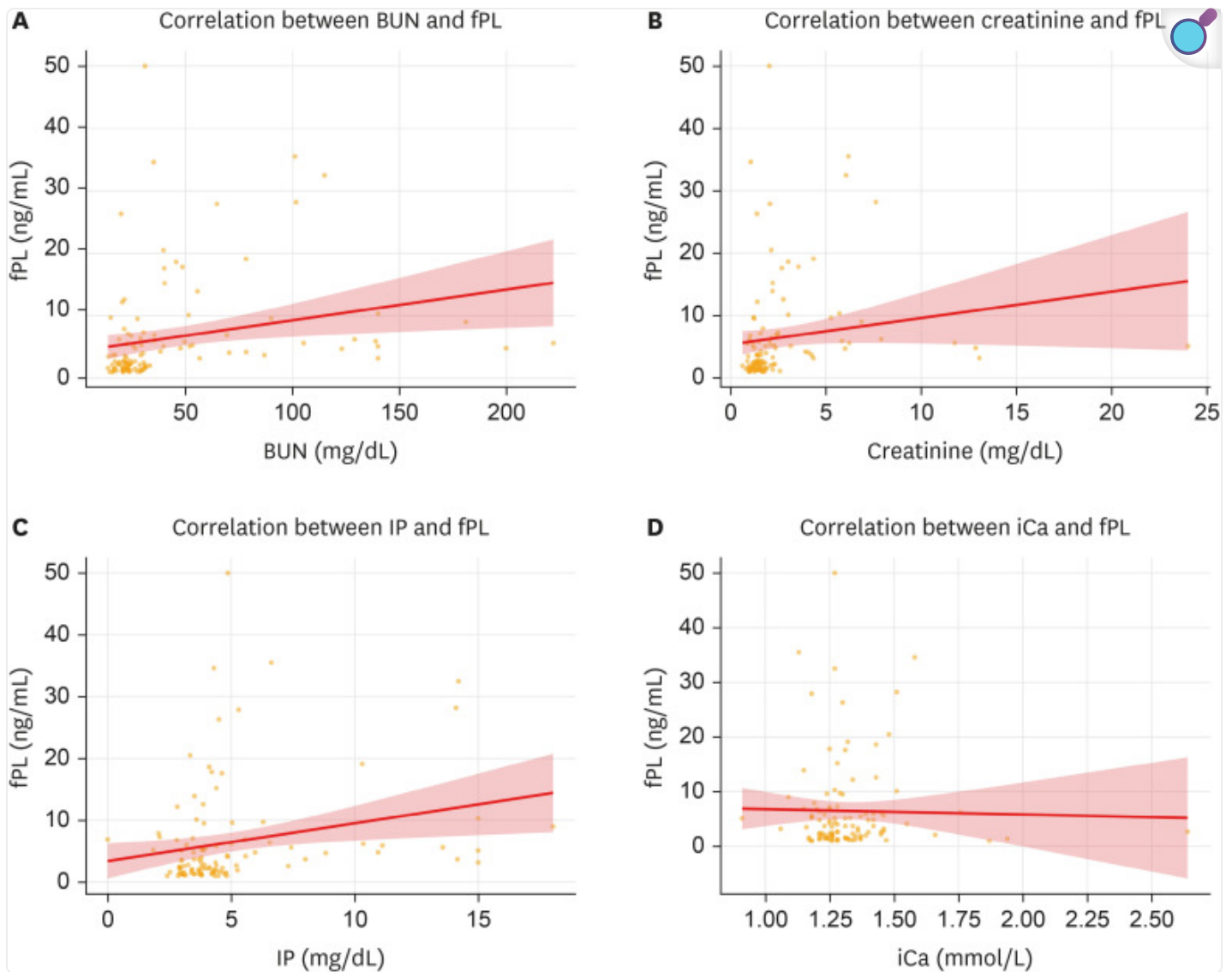
Post-hoc pairwise comparisons revealed that USG values in CKD stages 1 to 4 were significantly lower than those in the normal group (all $p < 0.001$). USG was also significantly lower for stages 3 and 4 than for stages 1 and 2 (all $p < 0.001$). No significant difference in USG was observed for CKD stages 3 and 4 ($p = 0.944$).

Relationship between renal biomarkers and fPL concentration

BUN

Serum fPL concentration showed a moderate positive association with BUN (Spearman's $\rho = 0.516$, $p < 0.001$), indicating that cats with higher BUN tended to have higher fPL concentrations ([Fig. 3](#)).

Fig. 3. Scatter plots depicting correlations between fPL concentrations and renal biomarkers in 112 cats: (A) BUN, (B) Creatinine, (C) IP, and (D) iCa.



[Open in a new tab](#)

Each point represents an individual cat. The red line indicates the line of best fit obtained from simple linear regression, and the shaded area represents the 95% confidence interval of the regression line. Positive correlations are observed between fPL and BUN, Creatinine, and IP concentrations. No significant correlation is found between iCa and fPL concentrations.

fPL, feline pancreatic lipase; BUN, blood urea nitrogen; IP, inorganic phosphorus; iCa, ionized calcium.

Creatinine

Serum fPL concentration showed a moderate positive association with serum creatinine (Spearman's $\rho = 0.459$, $p < 0.001$), indicating that cats with higher creatinine tended to have higher fPL concentrations.

IP

Serum fPL concentration showed a weak-to-moderate positive association with IP (Spearman's $\rho = 0.312$, $p < 0.001$).

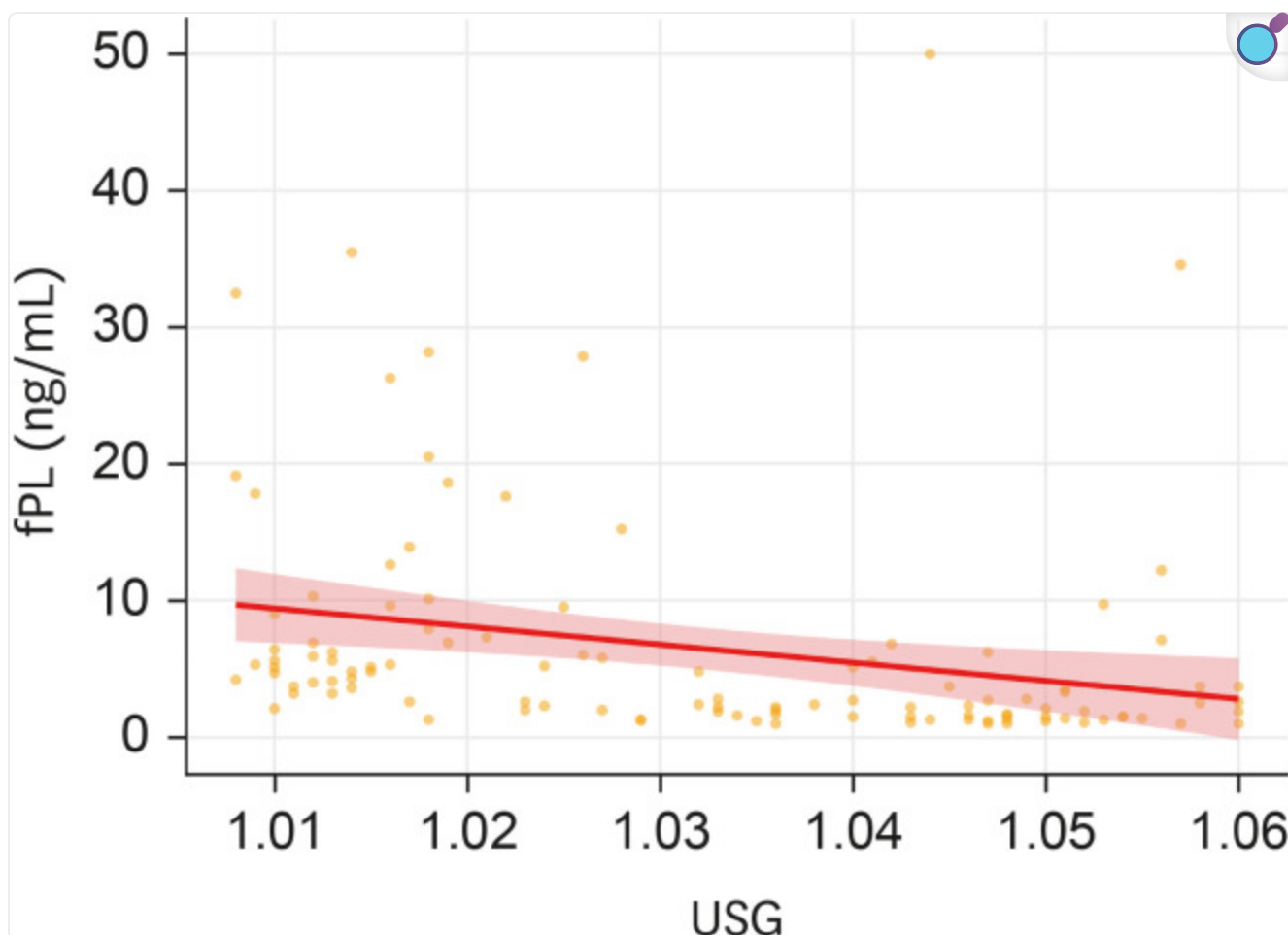
iCa

No meaningful association was identified between serum iCa and fPL concentration (Spearman's $\rho = 0.012$, $p = 0.898$).

USG

Serum fPL concentration showed a moderate negative association with USG (Spearman's $\rho = -0.502$, $p < 0.001$), indicating that cats with more dilute urine tended to have higher fPL concentrations ([Fig. 4](#)).

Fig. 4. Scatter plot depicting the correlation between USG and fPL concentrations in cats. A significant negative correlation is observed between USG and fPL concentrations, suggesting that lower USG values, indicative of impaired renal concentrating ability, are associated with higher fPL concentrations.



[Open in a new tab](#)

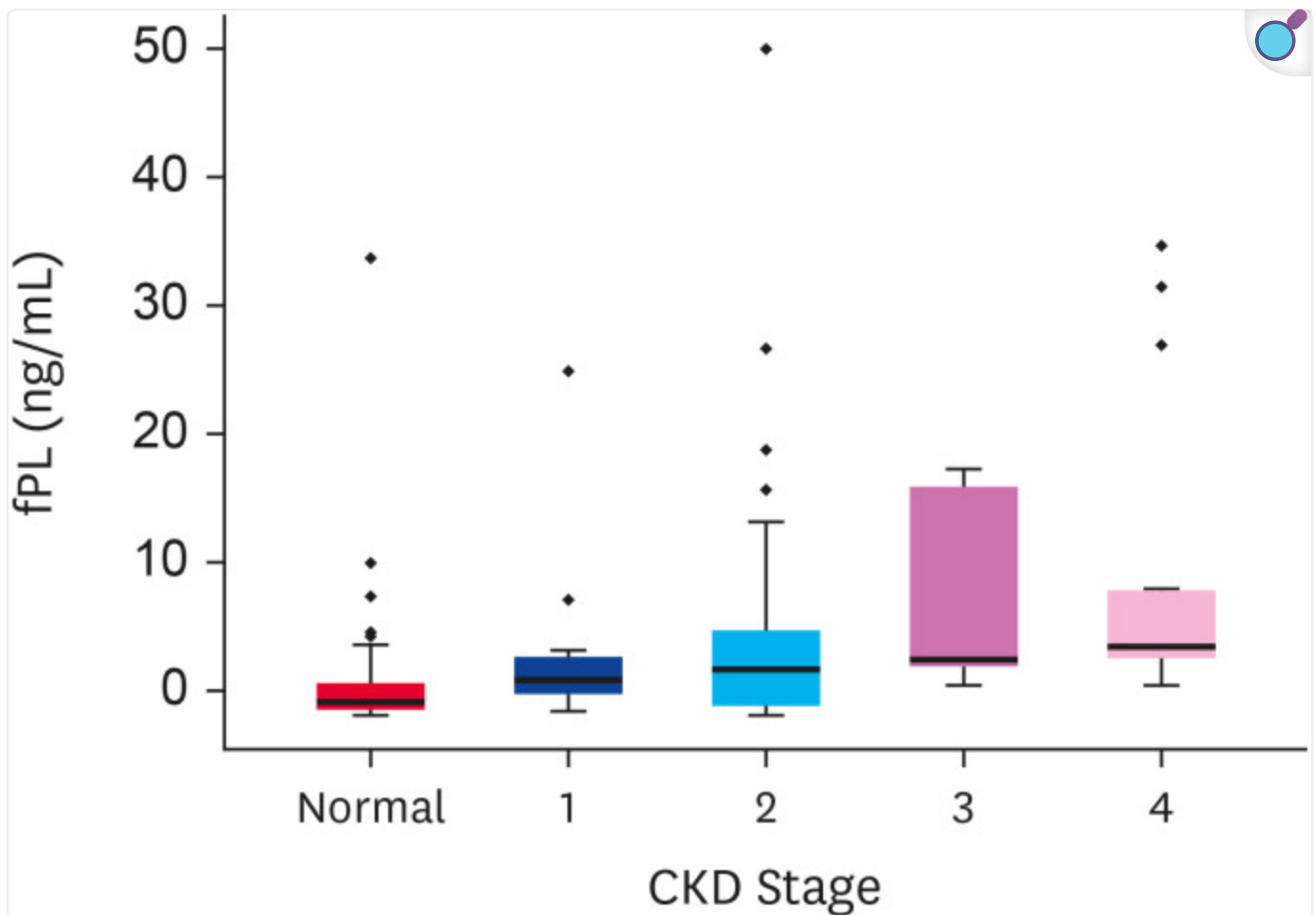
USG, urine specific gravity; fPL, feline pancreatic lipase.

Association between CKD stage and fPL

Serum fPL concentrations differed across CKD stages, with progressively higher values observed in cats with more advanced CKD ([Fig. 5](#)). Median fPL concentrations showed a progressive increase with advancing CKD stage: normal: 1.95 ng/mL (IQR, 1.4–3.325; 95% CI, 1.87–5.27); stage 1: 3.60

ng/mL (IQR, 2.6–5.3; 95% CI, 1.62–9.58); stage 2: 4.4 ng/mL (IQR, 1.7–7.2; 95% CI, 3.98–10.84); stage 3: 5.1 ng/mL (IQR, 4.1–17.8; 95% CI, 4.15–14.94); and stage 4: 6.05 ng/mL (IQR, 5.225–10.125; 95% CI, 5.40–18.35).

Fig. 5. Distribution of fPL concentrations across CKD stages in cats. Box-and-whisker plot depicting the distribution of fPL concentrations according to CKD IRIS stages in 112 cats. A progressive increase in fPL concentration is observed with advancing CKD stage. Markedly higher fPL concentrations are observed in cats with stages 3 and 4 of CKD.



[Open in a new tab](#)

fPL, feline pancreatic lipase; CKD, chronic kidney disease; IRIS, International Renal Interest Society.

The Kruskal–Wallis test revealed a significant difference in fPL concentrations across CKD stages ($H = 26.80, p < 0.001$). Post-hoc pairwise comparisons demonstrated that fPL concentrations were significantly higher in all CKD stages compared to the normal group (stage 1: $p = 0.04$; stage 2: $p = 0.04$; stage 3: $p < 0.01$; stage 4: $p < 0.01$). However, no significant differences were observed between adjacent CKD stages (stage 1 vs stage 2, $p = 0.95$; stage 2 vs stage 3, $p = 0.17$; stage 3 vs stage 4, $p = 0.48$). A significant difference was identified between early and advanced CKD, with stage 4 showing higher fPL concentrations than stage 1 ($p = 0.02$) and a borderline difference compared to stage 2 ($p = 0.05$).

DISCUSSION

This study assessed the association between serum fPL concentrations and CKD severity in cats, as well as their relationship with commonly used renal biomarkers. Serum fPL increased with worsening CKD stage and was associated with azotemia and reduced urine concentrating ability.

A progressive decline in USG was observed with advancing CKD stages, which were consistent with previous findings that associate CKD progression with impaired renal concentrating ability due to nephron loss [25,26]. However, no significant difference in USG was detected between CKD stages 3 and 4 ($p = 0.944$). This observation suggests that a critical threshold of nephron loss may have already been reached by stage 3, resulting in diminished concentrating capacity with minimal further decline as CKD advances to stage 4.

Serum fPL concentrations showed positive correlations with traditional renal biomarkers, including BUN, creatinine, and IP, all of which reflect impaired glomerular filtration and systemic azotemia. These associations suggest that declining renal function may influence measured fPL concentrations. The diagnosis of pancreatitis in cats remains inherently challenging, and an elevated fPL concentration alone is generally insufficient to establish clinically significant pancreatitis. Appropriate interpretation therefore requires integration of clinical signs and pancreatic ultrasonographic findings. Furthermore, pancreatic enzymes are known to increase when glomerular filtration is reduced, likely due to decreased renal clearance, which further complicates the interpretation of fPL results in cats with CKD.

A relevant experimental study in cats with induced CKD reported that serum feline trypsin-like immunoreactivity increased significantly with CKD, whereas serum fPL did not differ significantly from that in healthy controls and remained within reference intervals. This suggests that some pancreatic biomarkers may be variably influenced by renal dysfunction and highlights the need to interpret fPL results cautiously in the setting of azotemia [27].

In human medicine, the association between ESRD and AP is well-documented and attributed to multiple mechanisms. First, hypercalcemia is recognized as a leading cause of AP in advanced CKD, irrespective of dialysis status. Epidemiological data indicate that patients with CKD stage 5 who are not on dialysis and those undergoing dialysis are 4.59- and 1.65-fold, respectively, more likely to develop hypercalcemia-induced AP ($p < 0.01$) [19,20]. Second, the kidneys play a critical role in the degradation of gastrointestinal hormones such as cholecystokinin; impaired clearance can lead to persistent pancreatic stimulation and increased risk of AP. Third, intradialytic hypotension has been implicated in mesenteric ischemia, a known trigger for AP, with studies linking hypotension, high ultrafiltration rates, and elevated pancreatic enzyme concentrations [21]. Finally, peritoneal dialysis (PD) can increase the risk of pancreatitis through mechanisms including hypertriglyceridemia, local hypercalcemia, elevated intra-abdominal pressure, and PD-related peritonitis [22,23,28,29,30,31,32,33,34].

Taken together, our findings are most consistent with a passive (non-inflammatory) elevation of serum fPL concentrations associated with declining renal function. In the absence of standardized assessment of clinical signs and pancreatic imaging findings, elevations in fPL alone have inherent limitations for diagnosing pancreatitis in cats with CKD. In this context, our study provides a practical interpretive message for clinicians. Specifically, when mild-to-moderate increases in fPL are observed in cats with CKD, these findings should not be interpreted in isolation but rather evaluated in conjunction with compatible clinical signs and, when feasible, pancreatic-focused ultrasonography.

This study has some limitations. First, histopathological confirmation of pancreatic or renal lesions was not available, which precluded definitive diagnosis of the underlying disease. Second, pancreatic ultrasonographic findings and standardized clinical signs compatible with pancreatitis were not consistently documented, and therefore pancreatitis could not be defined using uniform diagnostic criteria. Third, hydration status at the time of presentation was not consistently or objectively quantified, and USG and some laboratory parameters may have been influenced by this factor. Finally, the retrospective cross-sectional design and inclusion of cats undergoing same-day testing may have introduced selection bias and limited assessment of temporal relationships.

In summary, serum fPL concentrations were associated with CKD severity and renal biomarkers in cats. These results indicate that reduced renal function may contribute to higher measured fPL concentrations and emphasize that fPL should not be used in isolation to diagnose clinically significant pancreatitis in cats with CKD.

Footnotes

Funding: This research was supported by the Regional Innovation System & Education (RISE) program through the Jeju RISE center, funded by the Ministry of Education (MOE) and the Jeju Special Self-Governing Province, Republic of Korea (2025-RISE-17-001).

Conflict of Interest: The authors declare no conflicts of interest.

Author Contributions:

Conceptualization: Kim J, Yun Y.

Data curation: Kim J.

Formal analysis: Kim J, Yun Y.

Funding acquisition: Yun Y.

Investigation: Kim J.

Methodology: Kim J, Yun Y.

Project administration: Kim J, Yun Y.

Resources: Kim J.

Software: Kim J.

Supervision: Yun Y.

Validation: Kim J.

Visualization: Kim J, Yun Y.

Writing - original draft: Kim J.

Writing - review & editing: Yun Y.

References

1. O'Neill DG, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Prevalence of disorders recorded in cats attending primary-care veterinary practices in England. *Vet J*. 2014;202(2):286–291. doi: 10.1016/j.tvjl.2014.08.004. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. White JD, Norris JM, Baral RM, Malik R. Naturally-occurring chronic renal disease in Australian cats: a prospective study of 184 cases. *Aust Vet J*. 2006;84(6):188–194. doi: 10.1111/j.1751-0813.2006.tb12796.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Marino CL, Lascelles BDX, Vaden SL, Gruen ME, Marks SL. Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited

for degenerative joint disease studies. *J Feline Med Surg*. 2014;16(6):465–472. doi: 10.1177/1098612X13511446. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

4. Lulich JP, Osborne CA, O'Brien TD, Polzin DJ. Feline renal failure: questions, answers, questions. *Compend Contin Educ Pract Vet*. 1992;14:127–152. [[Google Scholar](#)]

5. O'Neill DG, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Longevity and mortality of cats attending primary care veterinary practices in England. *J Feline Med Surg*. 2015;17(2):125–133. doi: 10.1177/1098612X14536176. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

6. Bartlett PC, Van Buren JW, Bartlett AD, Zhou C. Case-control study of risk factors associated with feline and canine chronic kidney disease. *Vet Med Int*. 2010;2010:957570. doi: 10.4061/2010/957570. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

7. Greene JP, Lefebvre SL, Wang M, Yang M, Lund EM, Polzin DJ. Risk factors associated with the development of chronic kidney disease in cats evaluated at primary care veterinary hospitals. *J Am Vet Med Assoc*. 2014;244(3):320–327. doi: 10.2460/javma.244.3.320. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

8. Hughes KL, Slater MR, Geller S, Burkholder WJ, Fitzgerald C. Diet and lifestyle variables as risk factors for chronic renal failure in pet cats. *Prev Vet Med*. 2002;55(1):1–15. doi: 10.1016/s0167-5877(02)00088-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

9. Jepson RE, Brodbelt D, Vallance C, Syme HM, Elliott J. Evaluation of predictors of the development of azotemia in cats. *J Vet Intern Med*. 2009;23(4):806–813. doi: 10.1111/j.1939-1676.2009.0339.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

10. Sparkes AH, Caney S, Chalhoub S, Elliott J, Finch N, Gajanayake I, et al. ISFM consensus guidelines on the diagnosis and management of feline chronic kidney disease. *J Feline Med Surg*. 2016;18(3):219–239. doi: 10.1177/1098612X16631234. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

11. De Cock HEV, Forman MA, Farver TB, Marks SL. Prevalence and histopathologic characteristics of pancreatitis in cats. *Vet Pathol*. 2007;44(1):39–49. doi: 10.1354/vp.44-1-39. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

12. Oppliger S, Hartnack S, Reusch CE, Kook PH. Agreement of serum feline pancreas-specific lipase and colorimetric lipase assays with pancreatic ultrasonographic findings in cats with

suspicion of pancreatitis: 161 cases (2008–2012) J Am Vet Med Assoc. 2014;244(9):1060–1065. doi: 10.2460/javma.244.9.1060. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

13. Rinderknecht H. In: The Pancreas: Biology, Pathobiology and Disease. 2nd ed. Go VLW, editor. Raven Press; 1993. Pancreatic secretory enzymes; pp. 219–251. [\[Google Scholar ↗\]](#)

14. Schnauß F, Hanisch F, Burgener IA. Diagnosis of feline pancreatitis with SNAP fPL and Spec fPL. J Feline Med Surg. 2019;21(8):700–707. doi: 10.1177/1098612X18796624. [\[DOI ↗\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

15. Forman MA, Marks SL, De Cock HE, Hergesell EJ, Wisner ER, Baker TW, et al. Evaluation of serum feline pancreatic lipase immunoreactivity and helical computed tomography versus conventional testing for the diagnosis of feline pancreatitis. J Vet Intern Med. 2004;18(6):807–815. doi: 10.1892/0891-6640(2004)18<807:eosfpl>2.0.co;2. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

16. Baggenstoss AH. The pancreas in uremia; a histopathologic study. Am J Pathol. 1948;24(5):1003–1017. [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

17. Avram MM. High prevalence of pancreatic disease in chronic renal failure. Nephron J. 1977;18(1):68–71. [\[Google Scholar ↗\]](#)

18. Araki T, Ueda M, Ogawa K, Tsuji T. Histological pancreatitis in end-stage renal disease. Int J Pancreatol. 1992;12(3):263–269. doi: 10.1007/BF02924366. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

19. Vaziri ND, Dure-Smith B, Miller R, Mirahmadi M. Pancreatic pathology in chronic dialysis patients--an autopsy study of 78 cases. Nephron J. 1987;46(4):347–349. [\[Google Scholar ↗\]](#)

20. Hou SW, Lee YK, Hsu CY, Lee CC, Su YC. Increased risk of acute pancreatitis in patients with chronic hemodialysis: a 4-year follow-up study. PLoS One. 2013;8(8):e71801. doi: 10.1371/journal.pone.0071801. [\[DOI ↗\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

21. Lankisch PG, Weber-Dany B, Maisonneuve P, Lowenfels AB. Frequency and severity of acute pancreatitis in chronic dialysis patients. Nephrol Dial Transplant. 2008;23(4):1401–1405. doi: 10.1093/ndt/gfm769. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

22. Chen HJ, Wang JJ, Tsay WI, Her SH, Lin CH, Chien CC. Epidemiology and outcome of acute pancreatitis in end-stage renal disease dialysis patients: a 10-year national cohort study. Nephrol Dial Transplant. 2017;32(10):1731–1736. doi: 10.1093/ndt/gfw400. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

23. Owyang C, Miller LJ, DiMagno EP, Mitchell JC, 3rd, Go VL. Pancreatic exocrine function in severe human chronic renal failure. *Gut*. 1982;23(5):357–361. doi: 10.1136/gut.23.5.357. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
24. Jaturanratsamee K, Choisunirachon N, Soontornvipart K, Darawiroj D, Srisowanna N, Thanaboonnipat C. Ultrasonographic kidney length-to-abdominal aortic diameter for the diagnosis of feline chronic kidney disease: a preliminary study. *Vet World*. 2023;16(5):1114–1121. doi: 10.14202/vetworld.2023.1114-1121. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Elliott J, Barber PJ. Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract*. 1998;39(2):78–85. doi: 10.1111/j.1748-5827.1998.tb03598.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Elliott J, Syme HM, Reubens E, Markwell PJ. Assessment of acid-base status of cats with naturally occurring chronic renal failure. *J Small Anim Pract*. 2003;44(2):65–70. doi: 10.1111/j.1748-5827.2003.tb00122.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Xenoulis PG, Moraiti KTR, R Finco D, Suchodolski JS, Steiner JM. Serum feline pancreatic lipase immunoreactivity and trypsin-like immunoreactivity concentrations in cats with experimentally induced chronic kidney disease. *J Vet Intern Med*. 2021;35(6):2821–2827. doi: 10.1111/jvim.16296. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
28. Barbara M, Tsen A, Rosenkranz L. Acute pancreatitis in chronic dialysis patients. *Pancreas*. 2018;47(8):946–951. doi: 10.1097/MPA.0000000000001119. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Sakorafas GH, Tsiotos GG, Sarr MG. Ischemia/reperfusion-induced pancreatitis. *Dig Surg*. 2000;17(1):3–14. doi: 10.1159/000018793. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
30. Chen YH, Yang WC, Wang FM, Tarng DC, Chen JY, Ng YY, et al. Risk factors associated with elevated serum pancreatic amylase levels during hemodialysis. *Hemodial Int*. 2011;15(1):79–86. doi: 10.1111/j.1542-4758.2010.00519.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
31. Owyang C, Miller LJ, DiMagno EP, Brennan LA, Jr, Go VL. Gastrointestinal hormone profile in renal insufficiency. *Mayo Clin Proc*. 1979;54(12):769–773. [[PubMed](#)] [[Google Scholar](#)]

32. Bruno MJ, van Westerloo DJ, van Dorp WT, Dekker W, Ferwerda J, Tytgat GN, et al. Acute pancreatitis in peritoneal dialysis and haemodialysis: risk, clinical course, outcome, and possible aetiology. *Gut*. 2000;46(3):385–389. doi: 10.1136/gut.46.3.385. [[DOI ↗](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
33. Quraishi ER, Goel S, Gupta M, Catanzaro A, Zasuwa G, Divine G. Acute pancreatitis in patients on chronic peritoneal dialysis: an increased risk? *Am J Gastroenterol*. 2005;100(10):2288–2293. doi: 10.1111/j.1572-0241.2005.41646.x. [[DOI ↗](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
34. Kroner PT, Mareth K, Raimondo M, Lee DD, Alsaad A, Aslam N, et al. Acute pancreatitis in advanced chronic kidney disease and kidney transplant recipients: results of a US nationwide analysis. *Mayo Clin Proc Innov Qual Outcomes*. 2019;3(2):160–168. doi: 10.1016/j.mayocpiqo.2019.03.006. [[DOI ↗](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
-

Articles from Journal of Veterinary Science are provided here courtesy of **The Korean Society of Veterinary Science**